

YANG YE

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EXPERIENCES

- **Associate Research Scientist** *2025.08 - present*
Center for Infectious Disease Modeling and Analysis
Yale School of Public Health
- **Affiliated Faculty** *2026.02 - present*
Yale Institute for Global Health
- **Postdoctoral Associate** *2023.08 - 2025.07*
Center for Infectious Disease Modeling and Analysis
Yale School of Public Health
Advisor: Alison Galvani
- **Visiting Scholar** *2021.07 - 2021.09*
UNC Project-China

EDUCATION

- **City University of Hong Kong** *2019.09 - 2023.06*
Doctor of Philosophy in Data Science
School of Data Science
Supervisor: Qingpeng Zhang
- **Huazhong University of Science and Technology** *2015.09 - 2019.06*
Bachelor of Management in Logistics Management
School of Artificial Intelligence and Automation

RESEARCH INTERESTS

Mathematical Epidemiology, Health Policy, Complex Networks

PUBLICATIONS

- **Ye, Y.**, Pandey, A., Fitzpatrick, M. C., & Galvani, A. P. (2026). US savings from international reference pricing for pharmaceuticals crucial to women's health. *The Lancet Obstetrics, Gynaecology, & Women's Health*, 2(6), e490-e491.
- Wells, C. R., Pandey, A., **Ye, Y.**, Bawden, C., Giglio, R., Wong, C., ... & Galvani, A. P. (2026). The health and economic repercussions of declining MMR coverage in the United States. *Proceedings of the National Academy of Sciences*, 123(17), e2605971123.
- **Ye, Y.**, Pandey, A., Fitzpatrick, M. C., Potter-Schwartz, L., Bawden, C., Bilori, B., ... & Galvani, A. P. (2026). Estimating US savings on outpatient prescription pharmaceuticals from international reference pricing. *Proceedings of the National Academy of Sciences*, 123(2), e2520871122 (Press Highlights).
- Pandey, A., **Ye, Y.**, Bawden, C., Singer, B. H., & Galvani, A. P. (2025). Quantifying the mortality and morbidity impact of medicaid work requirements: a modeling study. *The Lancet Regional Health - Americas*, 51, 101232.
- **Ye, Y.**, Pandey, A., Bawden, C., Sumsuzzman, D. M., Rajput, R., Shoukat, A., ... & Galvani, A. P. (2025). Integrating artificial intelligence with mechanistic epidemiological modeling: a scoping review of opportunities and challenges. *Nature Communications*, 16(1), 581 (Editors' Highlights).

- Sumsuzzman, D. M., **Ye, Y.**, Wang, Z., Pandey, A., Langley, J. M., Galvani, A. P., & Moghadas, S. M. (2025). Impact of disease severity, age, sex, comorbidity, and vaccination on secondary attack rates of SARS-CoV-2: a global systematic review and meta-analysis. *BMC Infectious Diseases*, 25(1), 215.
- Pandey, A., **Ye, Y.**, Wells, C. R., Singer, B. H., & Galvani, A. P. (2024). Estimating the lives that could be saved by expanded access to weight-loss drugs. *Proceedings of the National Academy of Sciences*, 121(43), e2412872121.
- **Ye, Y.**, & Zhang, Q. (2024). The futility of economic sanctions in a globalized and interdependent world: a data-driven game theoretical study. *Humanities and Social Sciences Communications*, 11(1), 1-10.
- **Ye, Y.**, Cao, Z., Dajun Zeng, D., & Zhang, Q. (2024). Supply chain loss from easing COVID-19 restrictions: an evolutionary economic-epidemiological modelling study. *Proceedings of the Royal Society A*, 480(2283), 20230803 (Cover feature).
- Jing, F., **Ye, Y. (Joint First Author)**, Zhou, Y., Ni, Y., Yan, X., Lu, Y., ... & Tang, W. (2023). Identification of key influencers for secondary distribution of HIV self-testing kits among Chinese men who have sex with men: development of an ensemble machine learning approach. *Journal of Medical Internet Research*, 25, e37719.
- Zhang, Q., **Ye, Y. (Joint First Author)**, & Zeng, D. D. (2023). Travel restrictions cannot prevent the introduction of new COVID variants. *Journal of Travel Medicine*, taad066.
- Cheng, W., Zhou, H., **Ye, Y.**, Chen, Y., Jing, F., Cao, Z., Zeng, D. D., & Zhang, Q. (2023). Future Trajectory of Respiratory Infections Following the COVID-19 Pandemic in Hong Kong. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 33(1), 013124.
- Liu, W., Pan, J., Zhang, X., Gong, X., **Ye, Y.**, Zhao, X., et al. (2023). Cross-platform product matching based on entity alignment of knowledge graph with RAEA model. *World Wide Web*. 10.1007/s11280-022-01134-y.
- **Ye, Y.**, Zhang, Q., Wei, X., Cao, Z., Yuan, H. Y., & Zeng, D. D. (2022). Equitable access to COVID-19 vaccines makes a life-saving difference to all countries. *Nature Human Behaviour*, 6(2), 207-216 (Cover feature).
- Jing, F., **Ye, Y. (Joint First Author)**, Zhou, Y., Zhou, H., Xu, Z., Lu, Y., ... & Wu, D. (2022). Modelling the geographical spread of HIV among MSM in Guangdong, China: a metapopulation model considering the impact of pre-exposure prophylaxis. *Philosophical Transactions of the Royal Society A*, 380(2214), 20210126.
- **Ye, Y.**, Ni, K., Jing, F., Zhou, Y., Tang, W., & Zhang, Q. (2022). Model-Informed Targeted Network Interventions on Social Networks Among Men Who Have Sex With Men in Zhuhai, China. *IEEE Transactions on Computational Social Systems*, 2022. 10.1109/TCSS.2022.3216756.
- **Ye, Y.**, Zhang, Q., Cao, Z., Chen, F. Y., Yan, H., Stanley, H. E., & Zeng, D. D. (2021). Impacts of Export Restrictions on the Global Personal Protective Equipment Trade Network During COVID-19. *Advanced Theory and Simulations*, 2100352 (Inner-cover featured).
- **Ye, Y.**, Zhang, Q., Cao, Z., & Zeng, D. D. (2021). Optimal vaccination program for two infectious diseases with cross-immunity spreading on a network. *EPL (Europhysics Letters)*, 133(4), 46001 (Featured as Editor's Pick).
- **Ye, Y.**, & Zhang, Q. (2021, August). Product Clustering Analysis Based on the Retail Product Knowledge Graph. In *Asia-Pacific Web (APWeb) and Web-Age Information Management (WAIM) Joint International Conference on Web and Big Data* (pp. 37-40). Springer, Singapore.
- **Ye, Y.**, Zhang, Q., Ruan, Z., Cao, Z., Xuan, Q., & Zeng, D. D. (2020). Effect of heterogeneous risk perception on information diffusion, behavior change, and disease transmission. *Physical Review E*, 102(4), 042314.

TEACHING EXPERIENCE

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- SDSC2004/GE2343: Data Visualization, Teaching Assistant *Spring, 2022*
 - SDSC2004: Data Visualization, Teaching Assistant *Spring, 2021*

SCHOLARSHIPS, HONORS AND AWARDS

- **Outstanding Academic Performance Award for Research Degree Students** *2022.09*
City University of Hong Kong
- **Research Tuition Scholarship** *2022.09*
City University of Hong Kong
- **Outstanding Academic Performance Award for Research Degree Students** *2021.09*
City University of Hong Kong
- **Research Tuition Scholarship** *2021.09*
City University of Hong Kong
- **Postgraduate Studentship** *2019.09*
City University of Hong Kong
- **China National Scholarship** (Highest undergraduate honor in China) *2017.11*
Huazhong University of Science and Technology
- **China National Scholarship** (Highest undergraduate honor in China) *2016.11*
Huazhong University of Science and Technology

PRESENTATIONS

- “Synergies Between Artificial Intelligence and Mechanistic Epidemiological Modeling”, invited lecture at the Infectious Disease Dynamics: Modeling and Computational Techniques workshop, Department of Mathematics, Birla Institute of Technology and Science, Pilani K K Birla Goa Campus, Goa, India, 2025.12
- “Equitable Access to COVID-19 Vaccines Makes a Life-Saving Difference to All Countries”, presentation delivered in the Centre for Complexity and Complex Networks at the City University of Hong Kong and the Department of Electronic and Information Engineering at the Hong Kong Polytechnic University, 2022.09
- “Equitable Access to COVID-19 Vaccines Makes a Life-Saving Difference to All Countries”, presentation delivered in NetSci2022 Satellite Forum, 2022.07
- “Product Clustering Analysis Based on the Retail Product Knowledge Graph”, presentation delivered in APWeb-WAIM 2021 International Workshops, 2021.08

REVIEWER OF JOURNALS

npj Digital Medicine, Philosophical Transactions of the Royal Society A, IEEE Transactions on Computational Social Systems, ACM Transactions on Management Information Systems, BMC Infectious Diseases, International Journal of Infectious Diseases

SELECTED MEDIA COVERAGE

- “Yale study: U.S. could save \$184 billion by aligning drug prices with peer nations.” Yale School of Public Health News, 2026. 01
- “Inteligencia artificial para prevenir y diagnosticar el dengue.” The Conversation, 2025.08
- “AI and epidemiology: Can combining the two predict pandemics?” The University of Florida Emerging Pathogens Institute, 2025.07
- “Access to weight-loss injectables remains limited for most needy in the US, Yale study finds.” Connecticut Public Radio, 2024. 11
- “Expanding Access to Weight-Loss Drugs Could Save Thousands of Lives A Year, Study Finds.” Yale Health & Medicine News, 2024. 10
- “Répartition des vaccins dans le monde.” CBC/Radio-Canada, 2022.10

- “The benefits of vaccinating the world are clear. The catch is the price tag.” The Washington Post, 2022.02
- “World better protected against Covid if rich nations donated 50% of vaccines to low-income countries.” The Guardian, 2022.02
- “COVID-19: 46% vaccine redistribution best for high- and low-income countries alike.” Nature Asia, 2022.02
- “A big win-win: Vaccine sharing could protect against future COVID-19 waves.” Gavi, the Vaccine Alliance, 2022.01
- “Mathematical model verifies a correct understanding of epidemic’s severity facilitates disease control.” CityU Research Stories, 2020.12